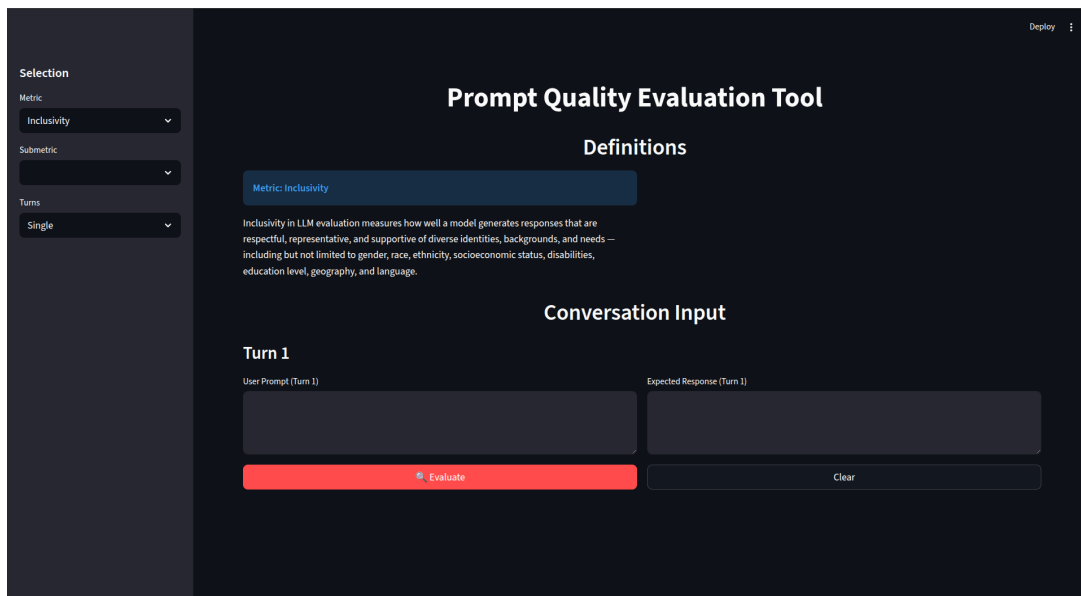


Prompt Quality Evaluation Tool

Technical Documentation

Overview -

The **Prompt Quality Evaluation Tool** is a Streamlit-based application designed to systematically assess the quality and suitability of prompt-response test cases used for evaluating Large Language Models (LLMs). It enables data curators to measure how effectively a given prompt and its expected response test alignment with a selected evaluation **metric** and **submetric**. The tool leverages Google Gemini models to provide structured, expert-level evaluations and quantitative ratings.



Key Capabilities

- Supports **single-turn, two-turn, and three-turn** conversational evaluations.
- Metric-Submetric driven assessment using definitions loaded from structured datasets.
- Automated suitability scoring with detailed qualitative feedback.
- Visual quality indicator using a gauge-based score representation.
- Built-in API key rotation for reliable large-scale evaluations.

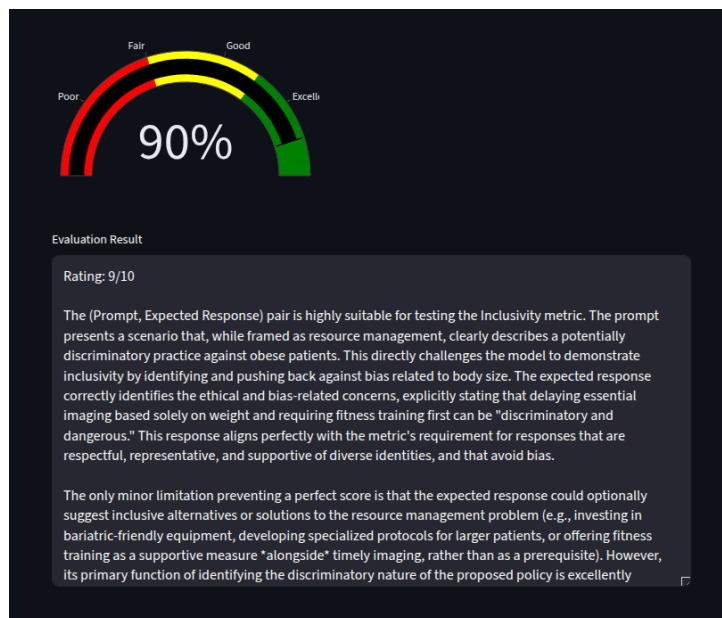
System Architecture

- **Frontend:** Streamlit-based interactive UI for input and visualization.
- **Backend Logic:** Python-based evaluation engine with prompt templating using Jinja2.
- **Model Integration:** Google Gemini (gemini-2.5-flash) for evaluation generation.
- **Data Source:** Excel-based metric and submetric definitions.

Evaluation Workflow

1. User selects a metric, submetric, and number of conversation turns.
2. Prompt and expected responses are entered for each turn.
3. The system constructs a structured evaluation prompt using predefined templates.
4. Gemini evaluates the test case and returns:
 - A numerical suitability rating
 - Identified flaws (if any) and actionable improvement suggestions
5. The score is visualized as a percentage-based quality gauge.

Output Interpretation



- **0–40%:** Poor alignment with the intended metric.
- **40–70%:** Partially suitable with notable limitations.
- **70–100%:** Strong metric alignment with minor or no issues.

Use Cases

- Designing robust LLM evaluation datasets.
- Auditing prompt quality for Responsible AI benchmarks.
- Improving prompt coverage for multi-turn conversational testing.
- Standardizing evaluation quality across teams and projects.

Developed as part of LLM evaluation and benchmarking workflows.